Supplementary Information

Dams mark drought vulnerability rather than buffer it, and storage cannot offset a falling water supply in Chile

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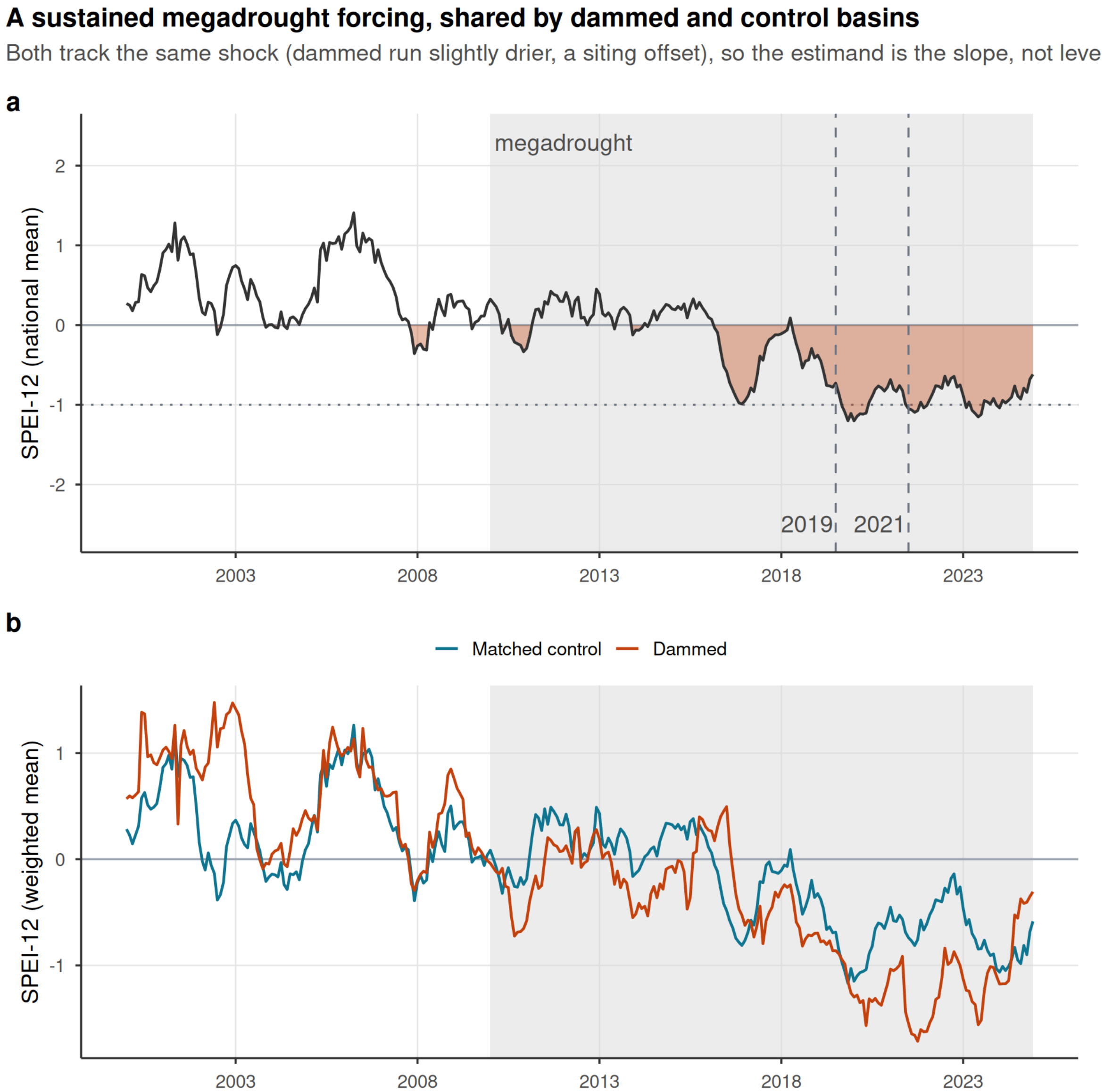
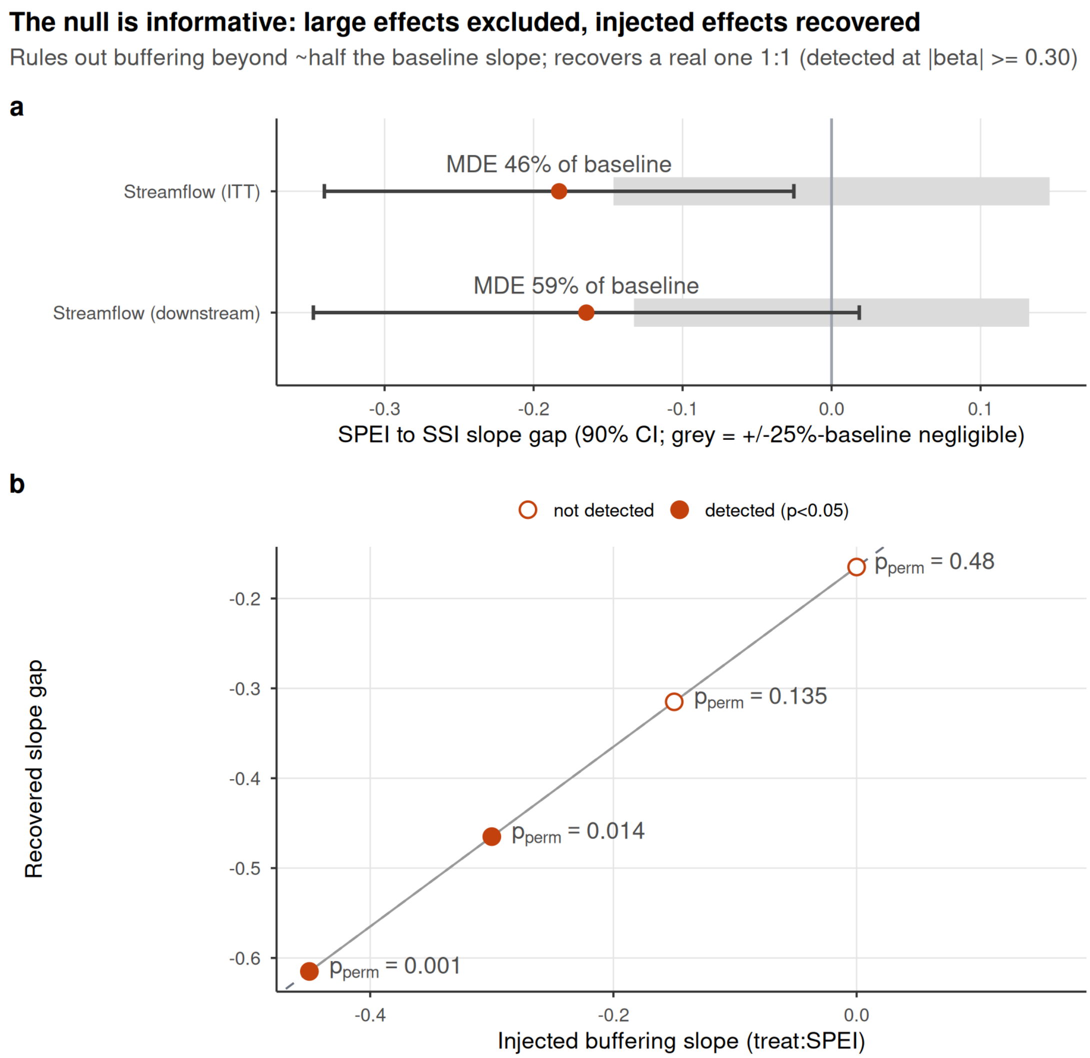
This Supplementary Information collects the supporting design, robustness, and diagnostic results referenced in the main text. Supplementary Figures S1–S3 document the drought forcing, the informativeness of the null, and the siting-confound decomposition; Supplementary Figures S4–S5 give the full study-area map and the matching diagnostics; Supplementary Figure S6 is the water-rights induced-demand test and Supplementary Figure S7 the barren-land ET-confound demonstration; Supplementary Tables S1–S6 report the equivalence bounds, placebo comparability, positive control, decomposition ladder, storage-band trends, and the aridity-overlap sensitivity; Supplementary Tables S7–S9 add the fully non-parametric overlap re-match, the winsorization-threshold sensitivity, and the spatial-autocorrelation assessment with spatially-restricted inference; Supplementary Table S10 bounds the groundwater-substitution confound with the DGA well hydrograph network; and Supplementary Table S11 tests reservoir-use heterogeneity (irrigation-only subset). All items are regenerated by the `targets` pipeline.

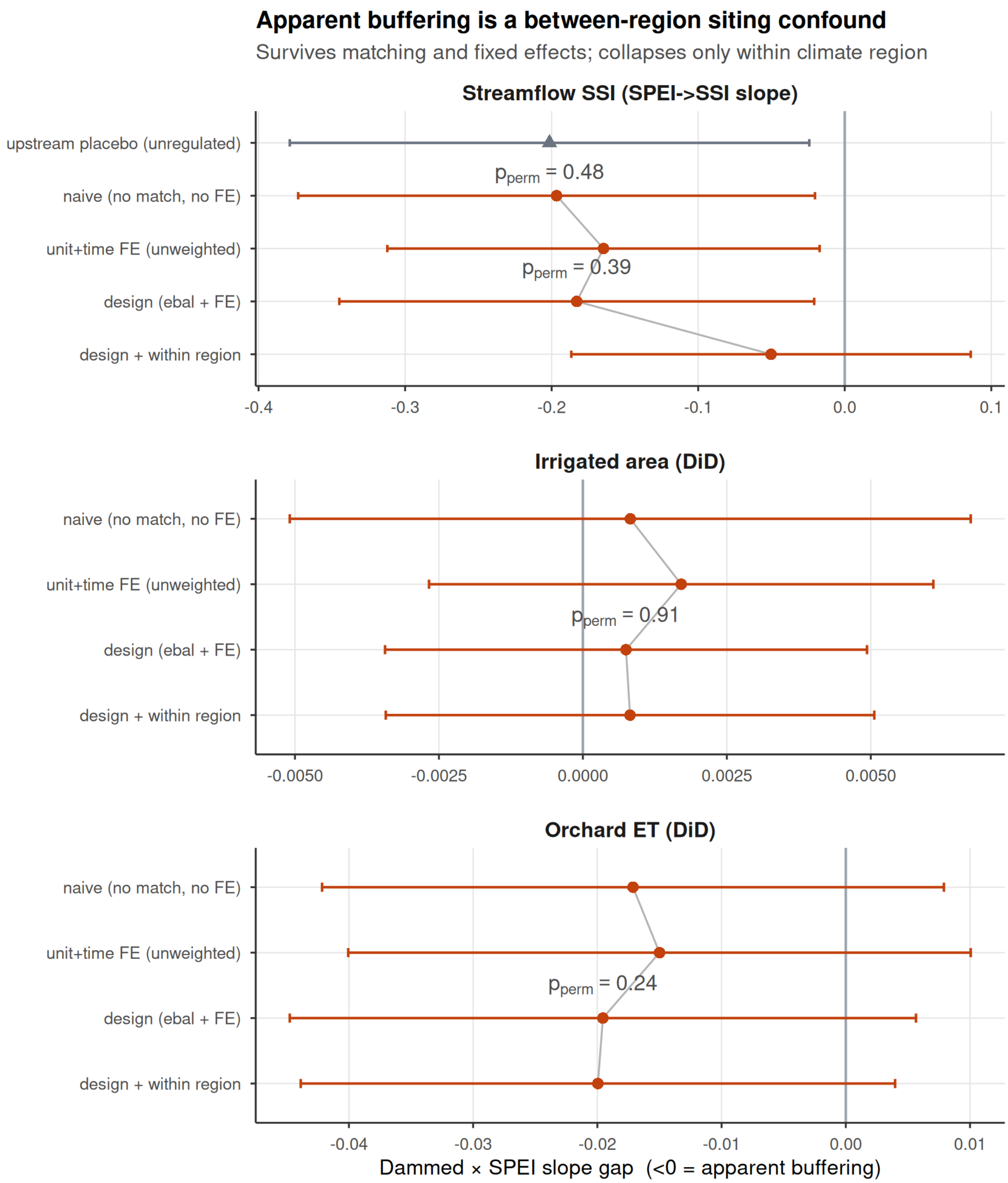
Supplementary Figures



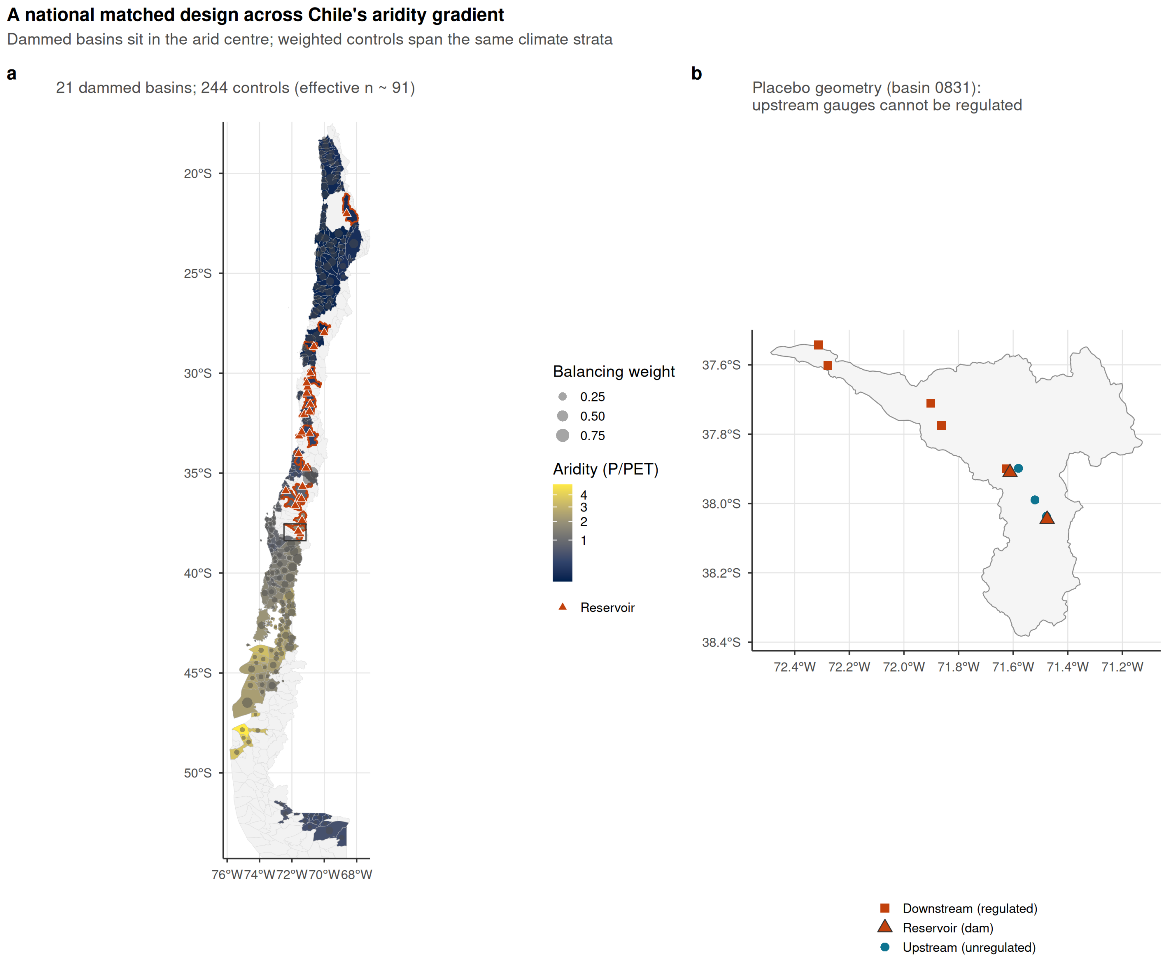
Supplementary Figure S1 | The drought forcing. (a) National mean SPEI-12 across the matched basins, 2000–2024; the negative (drought) excursion is filled and the post-2010 megadrought era shaded, with the 2019 and 2021 hyperdrought winters marked. The deficit is sustained and deep (below −1 for much of 2016–2024), giving the SPEI dose real dynamic range. (b) Entropy-balancing-weighted mean SPEI-12 for dammed versus matched-control basins: both co-move through the same megadrought (a common shock that year fixed effects absorb), with dammed basins running slightly drier, a siting offset that is precisely why the estimand is the differential deficit-to-impact slope rather than a level or calendar-time contrast.



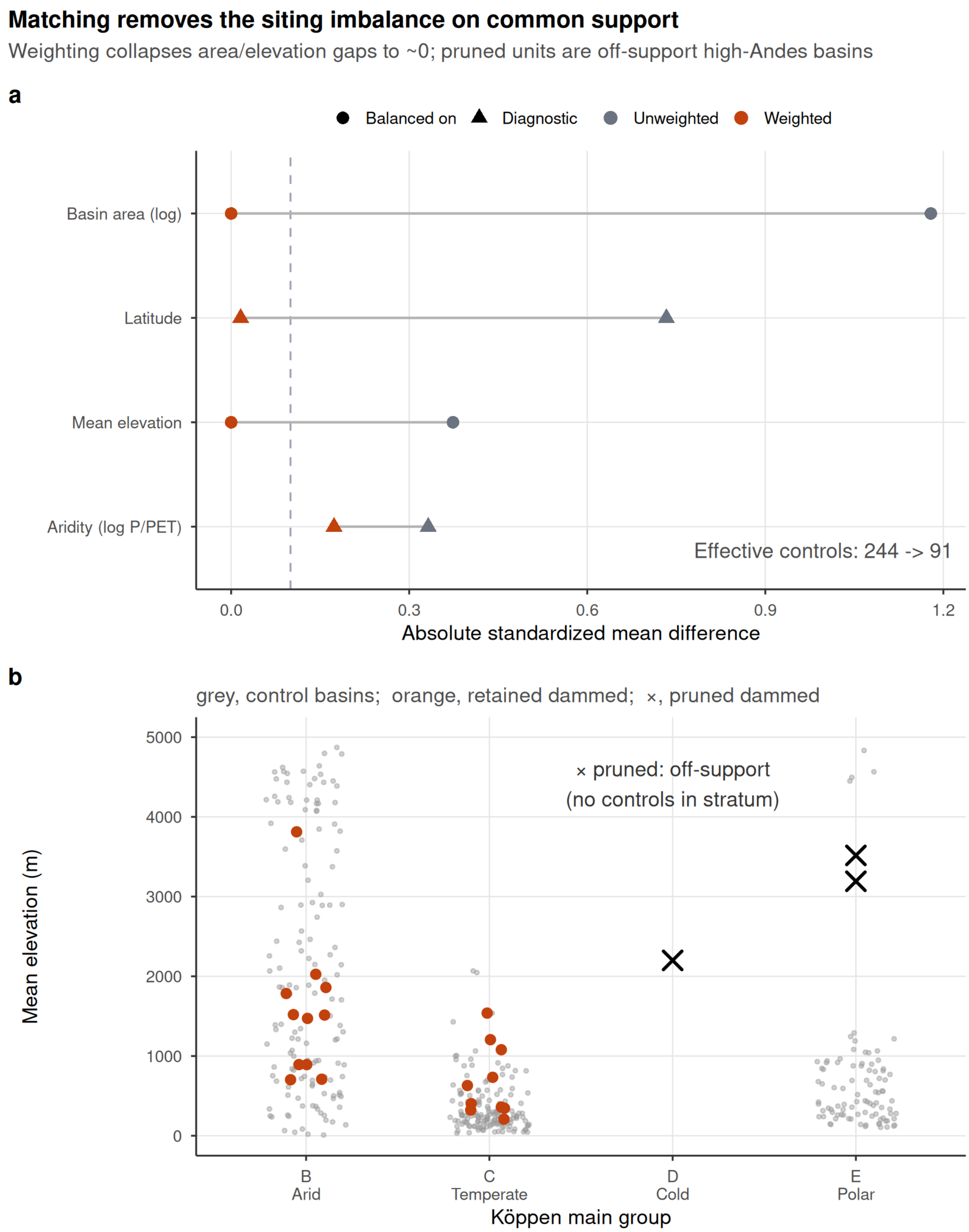
Supplementary Figure S2 | The null is informative, not underpowered blindness. (a) Equivalence / minimum-detectable-effect bounds for the two streamflow outcomes: the observed slope gap (point) and its 90% (TOST) interval against the ±25%-of-baseline negligible region (grey). The 90% intervals exceed the negligible region, so formal equivalence to zero is not claimed, but they exclude attenuations larger than the minimum detectable effect (45% of baseline transmission for the intent-to-treat design, 58% downstream). (b) Positive control: a known buffering slope injected into the treated downstream gauges is recovered one-to-one by the estimator (points on the dashed expected-recovery line), the no-injection case is correctly not flagged, and randomization inference detects the effect (filled points) once the injected slope reaches −0.30. The estimator would therefore reveal a large operational buffering effect if one existed.



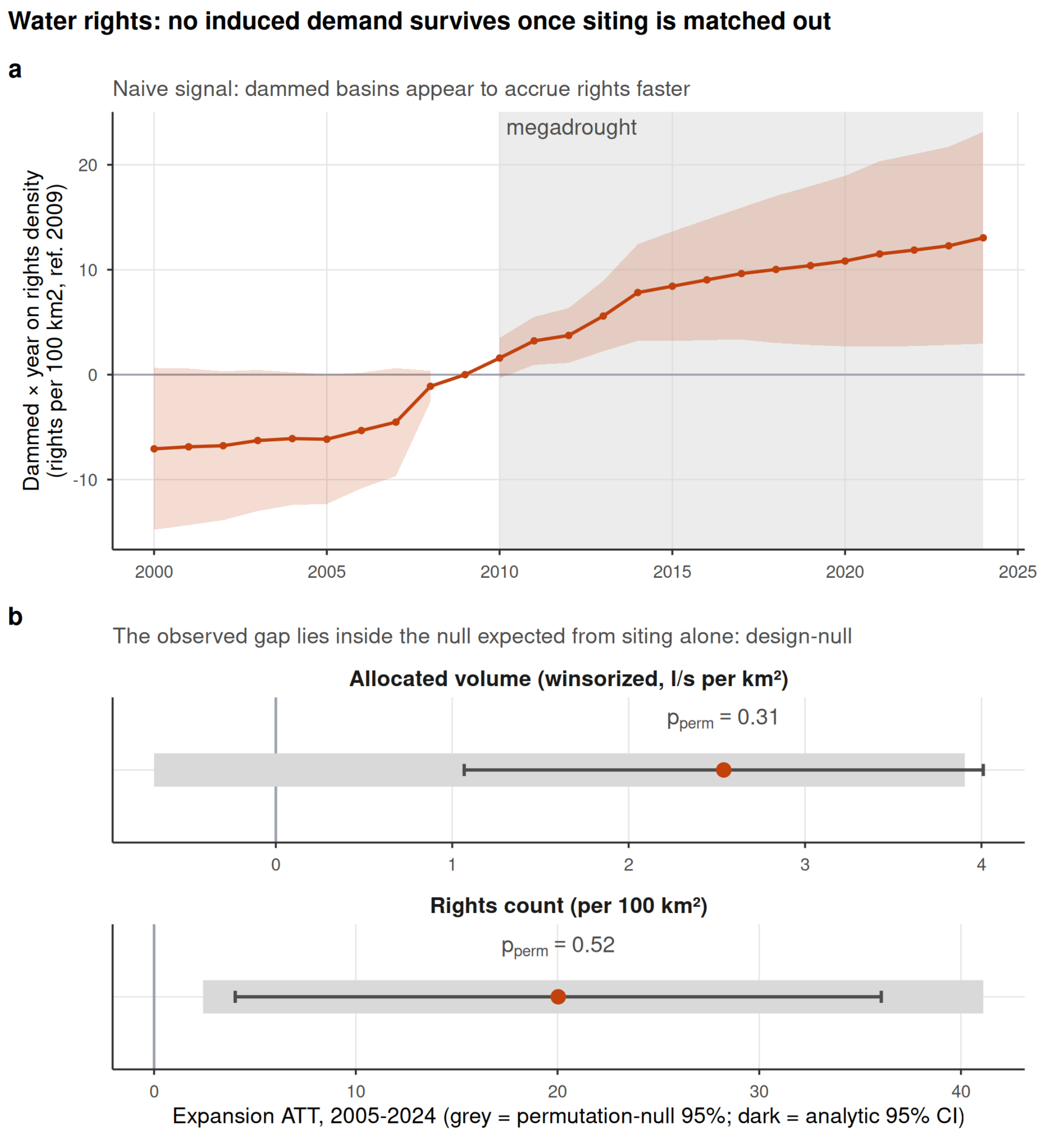
Supplementary Figure S3 | Siting-confound decomposition ladder. Dammed-versus-control transmission-slope gap (treat:SPEI; <0 = apparent buffering) estimated by increasingly stringent estimators, for the streamflow SSI slope (top) and the two difference-in-differences proxy outcomes; bars are 95% cluster-robust confidence intervals and the design and within-region rows carry the design's randomization-inference p. Rungs: (1) naive pooled (no matching, no fixed effects, what a conventional dammed-versus-undammed study reports); (2) unit and time fixed effects, unweighted; (3) the design estimator (entropy-balance weights + fixed effects); (4) design with the baseline SPEI-to-outcome slope allowed to vary within climate region. For streamflow the apparent buffering survives rungs (1) to (3) near -0.18 (permutation-null throughout), then collapses toward zero at rung (4), and is equally strong at the unregulated upstream placebo (grey triangle): the signature of a between-region aridity confound, not regulation. The proxy outcomes carry no signal at any rung.



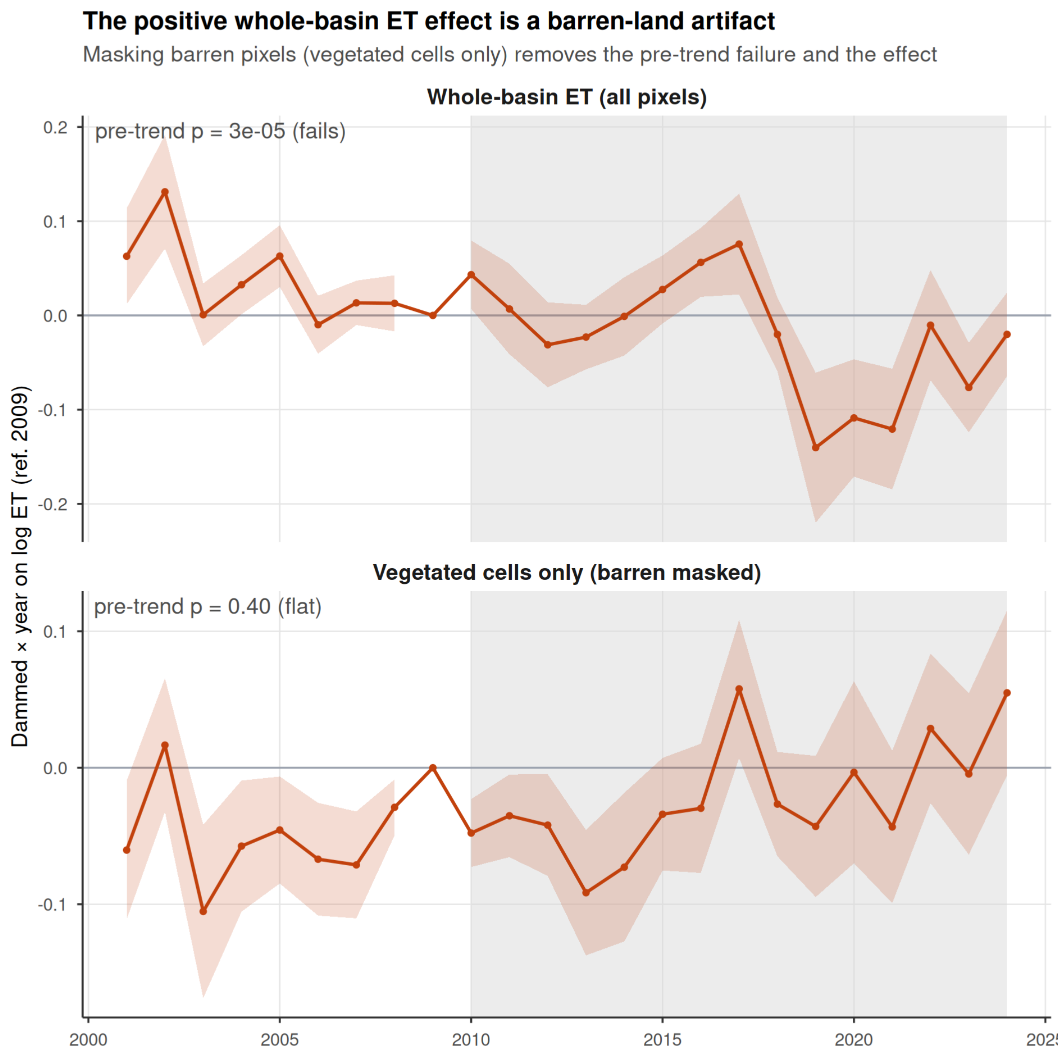
Supplementary Figure S4 | The national matched design across Chile's aridity gradient. (a) Matched sample of subcuencas, each filled by long-term aridity (P/PET, the north-south gradient the design conditions on); dammed (treated) basins are outlined in orange and marked with their reservoir (triangles), and each matched-control basin carries a centroid dot sized by its entropy-balancing weight, so the effective control sample (91) is visible against the nominal 244. Dammed basins concentrate in the arid centre and north; weighted controls span the same climate strata. (b) Within-basin placebo geometry for one representative dammed basin: the reservoir (triangle) with its downstream (regulated) gauges below the dam and upstream (unregulated) gauges above it.



Supplementary Figure S5 | Covariate balance and common support for the matched design. (a) Love plot of the absolute standardized mean difference between dammed and control basins before (unweighted) and after entropy balancing, for the two balanced covariates (log basin area, mean elevation) and two untargeted diagnostics (latitude, log aridity); the dashed line is the conventional 0.1 balance threshold. Weighting collapses the large siting imbalances to ~0, leaving a small residual aridity difference (0.17) that the doubly-robust outcome model absorbs, and reduces the effective control sample from 244 to 91. (b) Climate-stratum common support: mean elevation of control (grey) and dammed (orange, retained) basins within each Köppen main group; the three pruned dammed basins (x) fall in the cold (D) and polar (E) strata, whose controls lie far below them, so they are off-support and excluded rather than matched by extrapolation.



Supplementary Figure S6 | Water rights show no induced demand once siting is matched out. (a) Event study of cumulative consumptive water-rights density (dammed x year, ref. 2009; rights per 100 km2): the dammed-versus-control gap appears to widen through the megadrought, the naive induced-demand signal. (b) For both the rights count and the winsorized allocated volume, the expansion ATT (2005-2024; orange point, dark analytic 95% CI) is positive and its analytic interval excludes zero, yet it lies squarely inside the 95% band of the permutation null (grey), what random treatment assignment within climate x aridity strata already produces given the dammed basins' markedly higher aridity, so it is null under randomization inference (p_perm = 0.52 for counts, 0.31 for volume). A conventional matched analysis of this registry would report induced demand; the design-based inference reveals it as siting.



Supplementary Figure S7 | The positive whole-basin ET effect is a barren-land artifact. Event studies of the dammed-versus-control evapotranspiration response (dammed x year, ref. 2009) for (a) whole-basin ET over all pixels, which shows strongly non-parallel pre-trends (pre-trend slope p = 3x10^-5) and an apparent post-2010 divergence, and (b) ET restricted to vegetated irrigated (orchard) cells, i.e. with barren and salar pixels masked out. Masking the barren land removes both the pre-trend failure (p = 0.40) and the positive effect (the forcing-conditioned slope gap moves from +0.048 to -0.020), confirming that the exclusion of whole-basin ET from the main results is mechanistic (MOD16 ET is contaminated over sparsely vegetated barren surfaces), not a post-hoc drop of an inconvenient result.

Supplementary Tables

Supplementary Table S1 | Equivalence / minimum-detectable-effect bounds on the headline nulls. Observed slope gap, MDE (smallest effect detectable at 80% power from the permutation-null SD), the 90% (TOST) interval, and whether it lies within the negligible region (±25% of baseline). "baseline" is the untreated (control) SPEI→SSI / SPEI→impact transmission slope from the fixed-effects model. No outcome is formally equivalent to zero; the streamflow design rules out attenuations larger than ~46–58% of baseline transmission.

Outcome	Observed	MDE	90% CI	Equiv. to 0?	p_perm
streamflow SSI (ITT)	-0.183	0.268	[-0.34, -0.0253]	no	0.39
streamflow SSI (downstream)	-0.165	0.312	[-0.348, 0.0186]	no	0.48
irrigated area (DiD)	0.001	0.010	[-0.00506, 0.00656]	no	0.90
orchard ET (DiD)	-0.020	0.046	[-0.0463, 0.00718]	no	0.25

Supplementary Table S2 | Upstream/downstream placebo comparability. Upstream gauges are higher than downstream, and gauge elevation predicts the transmission slope among undammed controls (−0.21 per +1000 m), yet upstream and downstream treated gauges have nearly identical raw transmission slopes, so the equal up/down attenuation is not an elevation artifact.

quantity	value	detail
upstream gauges (elev, mean m)	1003.000	n=28; mean transmission 0.565
downstream gauges (elev, mean m)	459.000	n=40; mean transmission 0.599
control transmission slope vs elevation (per +1000 m)	-0.212	p = 0.00 (n=84 control gauges); ~0 => elevation does not drive transmission

Supplementary Table S3 | Positive control on the streamflow estimator. A known buffering slope (`beta_injected`; negative = buffering) is added to treated downstream units; the estimator recovers it (recovered = baseline −0.17 + beta), the no-effect case is not flagged, and randomization inference detects the effect once it exceeds the minimum detectable effect.

beta_injected	recovered	p_perm	detected
0.00	-0.165	0.480	FALSE
-0.15	-0.315	0.135	FALSE
-0.30	-0.465	0.014	TRUE
-0.45	-0.615	0.001	TRUE

Supplementary Table S4 | Siting-confound decomposition ladder (numeric). The dammed-versus-control transmission-slope gap at each rung of Supplementary Figure S3, for streamflow SSI (ITT) and the two difference-in-differences proxy outcomes, plus the within-basin upstream placebo. `perm_p` is the design's randomization-inference p for the design rung and the placebo.

Outcome	Rung	Slope gap	SE	p_perm
Streamflow SSI (SPEI->SSI slope)	naive (no match, no FE)	-0.197	0.090	-
Streamflow SSI (SPEI->SSI slope)	unit+time FE (unweighted)	-0.165	0.075	-
Streamflow SSI (SPEI->SSI slope)	design (ebal + FE)	-0.183	0.083	0.39
Streamflow SSI (SPEI->SSI slope)	design + within region	-0.050	0.070	-
Irrigated area (DiD)	naive (no match, no FE)	0.001	0.003	-
Irrigated area (DiD)	unit+time FE (unweighted)	0.002	0.002	-
Irrigated area (DiD)	design (ebal + FE)	0.001	0.002	0.91
Irrigated area (DiD)	design + within region	0.001	0.002	-
Orchard ET (DiD)	naive (no match, no FE)	-0.017	0.013	-
Orchard ET (DiD)	unit+time FE (unweighted)	-0.015	0.013	-
Orchard ET (DiD)	design (ebal + FE)	-0.020	0.013	0.24
Orchard ET (DiD)	design + within region	-0.020	0.012	-
Streamflow SSI (SPEI->SSI slope)	upstream placebo (unregulated)	-0.201	0.090	0.48

Supplementary Table S5 | Storage-band trends. Per-year trend slope of the cross-reservoir peak, trough, and peak-trough amplitude (percent of capacity per year; reservoir fixed effects, cluster-robust). Peak and trough both decline while the amplitude trend is not distinguishable from zero, consistent with a supply-side level decline rather than a loss of refill capacity.

Component	Trend (%/yr)	95% CI	p
peak	-1.281	[-2.21, -0.349]	0.013
trough	-1.220	[-1.72, -0.72]	0.000
amplitude	-0.061	[-1.13, 1.01]	0.912

Supplementary Table S6 | Aridity-overlap sensitivity. Key results re-estimated on the sub-sample of basins whose baseline log-aridity lies in the treated/control overlap band (all 21 treated basins and 158 of 244 controls fall inside it), to show the parametric aridity adjustment is not masking a true effect where common support is genuine. The streamflow transmission-slope gap is essentially unchanged, and the water-rights expansion ATT shrinks and now straddles zero.

Quantity	Full sample	Aridity overlap	Overlap 95% CI
Streamflow SPEI->SSI slope gap	-0.183	-0.174	-
Water-rights expansion ATT	20.036	11.880	[-5.62, 29.4]

Supplementary Table S7 | Fully non-parametric matching on the aridity-overlap subset. Every cross-sectional outcome re-estimated after re-fitting entropy balancing *within* the treated/control aridity-overlap band and using the weighting-only estimator (ebal-weighted difference in means, no outcome-model aridity term), so the estimate performs no parametric extrapolation across the aridity regime gap. Cropland-area expansion, orchard-ET buffering and the water-rights count remain null. The winsorized water-rights volume is the only outcome whose stratified-permutation p falls below 0.05 here; Supplementary Table S9 shows this reflects spatial autocorrelation, not a reservoir effect. The `perm p` column is the unit-level stratified randomization-inference p (Köppen × aridity tercile).

Outcome	ATT	95% CI (HC3)	p_perm	n_t	n_c
Cropland-area expansion (ha/km2)	-0.5860	[-2.08, 0.905]	0.328	21	158
Orchard ET buffering (dlogET/dSPEI)	-0.0182	[-0.0427, 0.00631]	0.291	18	30
Water-rights count (per 100 km2)	12.0000	[-4.36, 28.4]	0.420	21	136
Water-rights volume (l/s per km2)	1.8700	[0.164, 3.58]	0.005	21	136

Supplementary Table S8 | Sensitivity of the water-rights volume null to the winsorization threshold. The winsorized allocated-volume expansion ATT and its randomization-inference p, with individual consumptive flows capped at the 95th, 99th, 99.5th (the value used in the main text) and 99.9th percentiles of the consumptive-flow distribution. The analytic interval excludes zero at every threshold, but the effect is not significant under randomization inference at any threshold (permutation p from 0.39 to 0.21), so the volume null is not an artefact of the winsorization choice.

Winsor pctile	Volume ATT	95% CI	p_perm
95.0	1.837	[0.639, 3.03]	0.389
99.0	2.395	[0.956, 3.83]	0.343
99.5	2.539	[1.07, 4.01]	0.310
99.9	3.211	[1.56, 4.86]	0.209

Supplementary Table S9 | Spatial autocorrelation and spatially-restricted inference. Top: Moran's I of each outcome's ATT-model residuals (great-circle distances, five-nearest-neighbour row-standardized weights, 999 residual permutations). The water-rights outcomes and cropland area show significant positive spatial autocorrelation, which the unit-level stratified permutation does not preserve. Bottom: for the water-rights outcomes, the ATT with its spatially-naive (HC3) interval and stratified-permutation p alongside the cuenca-block-clustered interval and the spatial-block permutation p (treatment swapped at the cuenca level). The apparent water-rights volume effect on the overlap subset (stratified $p = 0.003$) is not significant once spatial dependence is respected (spatial-block $p = 0.077$; cuenca-clustered interval spans zero), and the count is null under both.

Outcome	n	Moran's I	Expected	p
Cropland-area expansion	265	0.181	-0.004	0.001
Water-rights count	216	0.599	-0.005	0.001
Water-rights volume	216	0.388	-0.005	0.001

Outcome	n	Moran's I	Expected	p
Orchard ET buffering	49	0.069	-0.021	0.198

Outcome	ATT	95% CI (HC3)	p (stratified)	95% CI (cuenca)	p (spatial block)
Water-rights volume (l/s per km2)	1.87	[0.16, 3.58]	0.005	[-0.33, 4.07]	0.077
Water-rights count (per 100 km2)	12.00	[-4.36, 28.36]	0.420	[-2.70, 26.70]	0.581

Supplementary Table S10 | Groundwater-substitution bound from the DGA well hydrograph network. Differential groundwater drawdown between dammed and matched-control basins over the megadrought (2010–2021). Each well's depletion rate is the OLS slope of Hampel-filtered depth to water on time (m yr^{-1}, positive = falling water table); per-basin rates are aggregated by the outlier-robust median (primary) and, for robustness, the mean. `dammed` / `control` are the weighted group-mean rates; `att_wo` is the entropy-balancing weighting-only ATT (with 95% CI and stratified randomization-inference `perm_p`) and `att_dr` the aridity-adjusted doubly-robust ATT. Coverage is 26 matched basins (13 dammed, 13 control, 213 wells). The differential is not significant in either specification, and the depletion residuals carry no spatial autocorrelation (`moran_I`, `moran_p`), so dammed basins show no faster groundwater drawdown of the kind a substitution confound would require.

Spec	Dammed (m/yr)	Control (m/yr)	ATT (WO)	95% CI	p_perm	ATT (DR)	Moran I	Moran p	wells
median (primary)	0.098	0.079	0.089	[-0.17, 0.35]	0.681	0.028	-0.069	0.752	213
mean (robustness)	0.289	0.076	0.283	[-0.11, 0.67]	0.328	0.486	-0.014	0.726	213

Supplementary Table S11 | Reservoir-use heterogeneity: irrigation-only robustness. The treated sample is dominated by irrigation reservoirs (18 of the 21 matched treated basins hold one; three are purely hydropower or potable supply). The headline streamflow SPEI-to-SSI slope gap (`treat:SPEI`, < 0 = apparent buffering) is re-estimated on all treated basins and on the irrigation-only subset (non-irrigation treated basins dropped, controls unchanged), with the design's stratified randomization-inference p. The slope gap and its permutation-null status are unchanged, so the null is not an artefact of pooling hydropower with irrigation reservoirs; `n_treat` columns give the treated basins with streamflow gauges in each case.

Quantity	All treated	p (all)	Irrigation only	p (irrig)	n_treat (all)	n_treat (irrig)
Streamflow slope gap (ITT)	-0.183	0.388	-0.192	0.432	17	16
Streamflow slope gap (upstream placebo)	-0.201	0.482	-0.221	0.456	15	14